# Lab 6: Analyzing the Effectiveness of Different Sterilization Techniques

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## Materials and Methods

Stainless steel 316L (SS) and alumina (A) samples were used as model biomaterials. Escherichia coli (E. coli) vitroids were used as the biological agent. Tryptic soy broth (TSB), BioTracker Bacterial FtsZ Live Cell Dye, phosphate-buffered saline (PBS), and 70% ethanol were used for bacterial culture and analysis. A total of 22 samples were prepared, including three replicates of each material (SS and alumina) for each sterilization method (autoclave, UV radiation, and ethanol), along with positive and negative controls.

Initial baseline images of all samples were captured using an optical microscope. To create a standardized bioburden, E. coli was diluted to a concentration of $1 \times 10^6$ bacteria/mL. Twenty samples were inoculated with the bacterial suspension and incubated for 30 minutes to allow bacterial adhesion. One untreated sample of each material served as a negative control.

Following incubation, samples were subjected to one of three sterilization methods: autoclaving, UV radiation, or ethanol treatment. The autoclave was set to a temperature and pressure of 121°C and 15 psi for 30 minutes. UV radiation was completed at 40 μW/cm² for 1 hour (total dose ≈ 144 mJ/cm²), and the ethanol treatment was completed with 70% ethanol for 20 minutes, followed by three PBS rinses. Three replicates of each material were used for each sterilization condition. Positive control samples (inoculated but not sterilized) and negative controls (uninoculated) were included for comparison.

After sterilization, all samples were imaged using optical microscopy to qualitatively assess bacterial presence and surface changes. Pre-inoculation, post-inoculation, and post-treatment images were compared. Bacterial viability was quantified using TSB turbidity measurements at 600 nm with a plate reader. Increased absorbance corresponded to higher bacterial growth. Samples were incubated with BioTracker Bacterial FtsZ Live Cell Dye (2 μg/mL) for 30 minutes at room temperature in the dark. Samples were then rinsed twice with PBS and imaged using a fluorescence microscope (excitation: 488 nm, emission: 510 nm). Live bacteria were identified by green fluorescence.

# Results

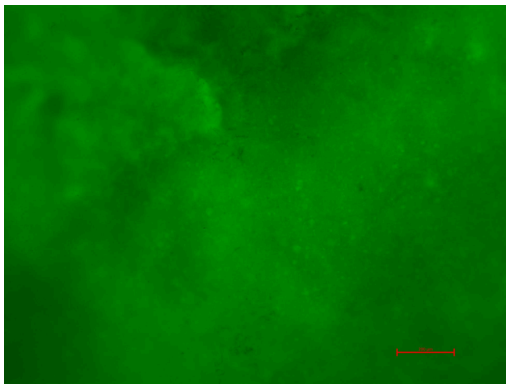


***Figure 1 - Alumina Autoclave Result [1]***

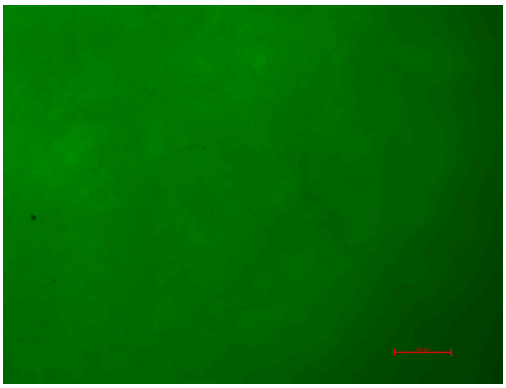


***Figure 2 - Alumina Ethanol Result [1]***

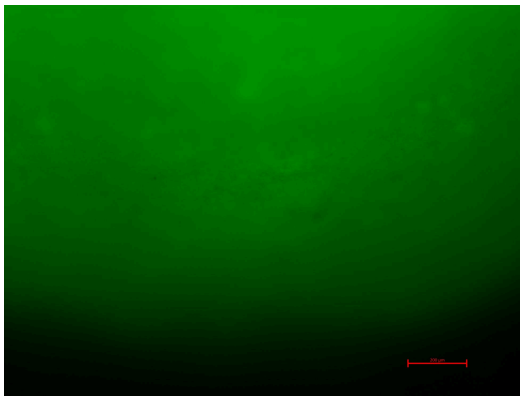


***Figure 3 - Alumina UV Result [1]***

***Table 1 - Summary of p-value and technique comparison***

| T-Test Comparison | P-Value | Statistical Significance (P< 0.05) |
|---|---|---|
| *Alumina/Stainless Steel* | 0.313 | No |
| *UV/ Positive Control* | 0.154 | No |
| *UV/ Negative Control* | 0.409 | No |
| *Ethanol/ Positive Control* | 0.158 | No |
| *Ethanol/ Negative Control* | 0.773 | No |
| *Autoclave/ Positive Control* | 0.154 | No |
| *Autoclave/ Negative Control* | 0.415 | No |

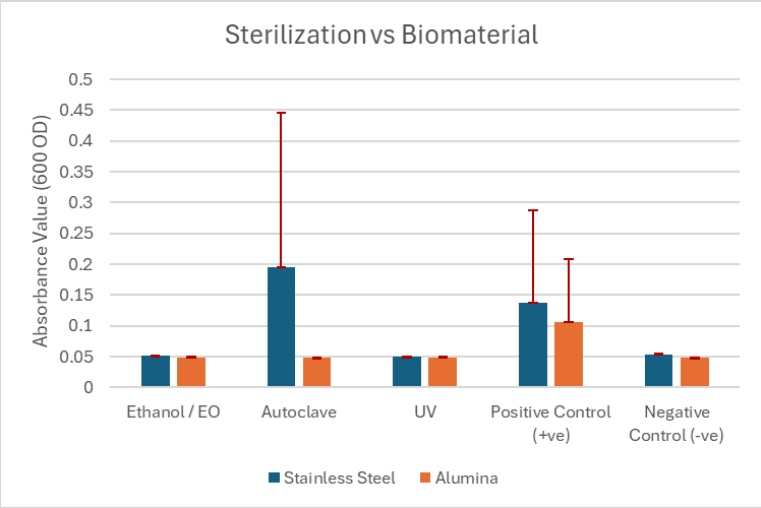



***Figure 4 - Absorbance Value Summary Chart***

# Discussion

To quantitatively evaluate if two materials (alumina and stainless steel) had statistically different results from either of the three sterilization techniques (autoclave, ethanol treatment and UV-exposure), a two-sample t-test was performed assuming equal variances. The results have been summarized in ***Table 1***. According to ***Table 1***, there was no significant difference between the positive, negative and E.Coli-treated samples for alumina and stainless steel for either of the three sterilization techniques used. However, the positive controls did not exhibit the expected high absorbance or visual turbidity, indicating that there was likely insufficient or no bacterial loading. Though the positive controls did show slightly higher absorbance values compared to the other conditions as seen in ***Figure 4***, the values are low and overlap with sterilized samples, indicating inconsistent bacterial loading. As a result, it is difficult to draw definitive conclusions from the statistical analysis, since the baseline condition of presence of

bacterial growth was not properly established. Since the positive controls demonstrated no bacterial loading, conclusions cannot be drawn regarding the effectiveness of the different sterilization methods. If we assume that bacterial loading did occur, the results would suggest that there is no statistical significance between the three sterilization techniques for either of the materials tested. This could be interpreted to suggest that all the methods were similarly effective, and their selection as a sterilization technique would depend on material/technology availability and associated costs. On the other hand, the lack of observed differences could be due to experimental error, including improper preparation of bacteria samples or inconsistencies with handling.

To qualitatively evaluate differences, fluorescence images from the Wednesday group were analyzed to determine if there were different results for alumina treated with E.Coli after each sterilization method. ***Figures 1*** through ***3*** summarize these visual observations. Results were consistent with the quantitative results; there were no significant differences observed between the images. However, minor variations were present in the images. For example, ***Figure 1*** has a slightly cloudy appearance, ***Figure 2*** has a couple of speckles on the lower left side, and ***Figure 3*** has a slight difference in contrast with the bottom half of the image being slightly darker. However, these visual differences do not indicate differences in presence of bacteria or sterilization effectiveness.

A major limitation of this experiment was the failure of bacteria culturing, as shown in the results of the positive controls. This was likely due to the E. coli medium being too diluted to support sufficient growth and show adequate results, contributing to the inconclusive results. Additionally, having to align the experimental design timeline with the lab schedule was another limitation experienced during this experiment. The initial experimental design involved allowing the E. coli to grow for 48 hours so that it would reach the stationary phase, but this had to be shortened to 24 hours due to scheduling limitations. This was a potential limitation that might have further decreased bacterial loading. To enhance the effectiveness of UV sterilization, a potential improvement could be to wipe surfaces with alcohol before completing UV sterilization. The mechanical action of wiping back and forth ensures that dust particles or other particles that obstruct UV rays are removed, allowing for full penetration and effectiveness of UV. Outside of UV sterilization, this practice of wiping surfaces is applicable for autoclave and ethanol treatments as well. This is because surface contaminants such as proteins or enzymes may enhance bacterial growth and interfere with sterilization processes [2].

# Acknowledgements

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# Supplemental Materials

ELEN here

# References

[1] “Google Drive folder,” *Google Drive*. [Online]. Available: https://drive.google.com/drive/folders/1HeRd79p2ZClYkk_GqQVnXkERY4ZoXkVN

[2] M. N. Panousi, G. J. Williams, S. Girdlestone, S. J. Hiom, and J. Y. Maillard, “Evaluation of alcohol wipes used during aseptic manufacturing,” *Letters in Applied Microbiology*, vol. 48, no. 5, pp. 648–651, 2009. [Online]. Available: https://doi.org/10.1111/j.1472-765X.2009.02574.x